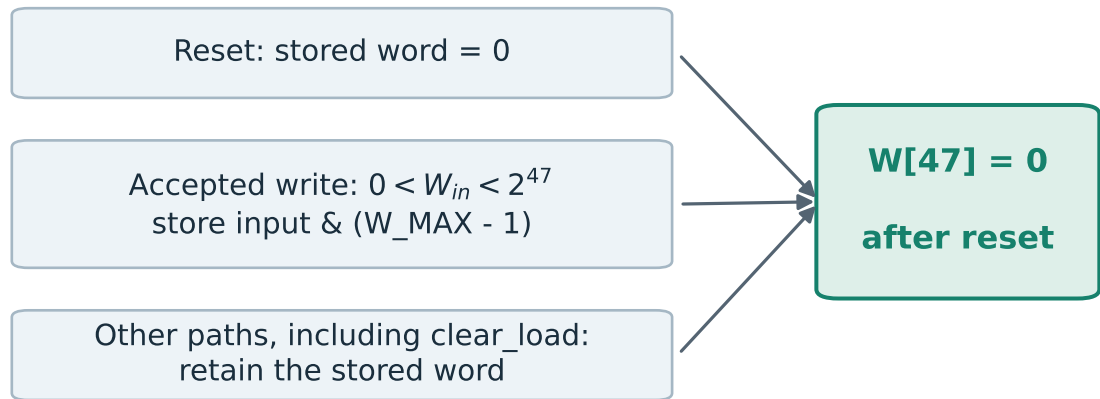


Weight-store MSB: proof, baseline witness, and verification

The accepted-write mask exposes an existing post-reset invariant; it preserves the 48-bit interface and accepted precision.

A Mathematical invariant



Low 47 bits are retained exactly; invalid-write rejection is unchanged.

B Repaired-baseline netlist evidence

1,278 FDRE

Actual bit47 instances: w_q_reg[*][*][47]



61,344 weight-value FFs = 18 x 71 x 48

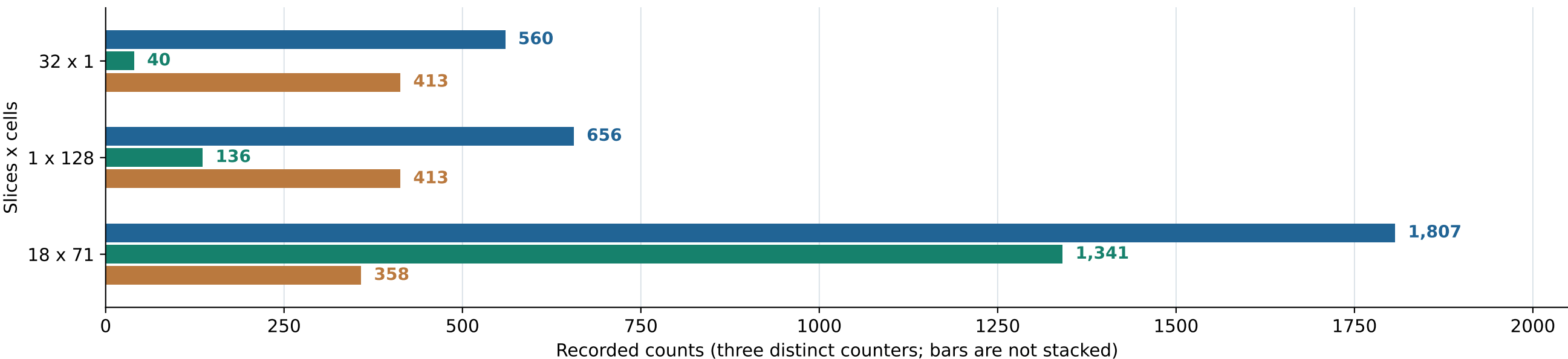
Green segment: the 1 / 48 high-bit share of this baseline.

EDIF line 4,757,998: w_q_reg[0][0][47] -> FDRE

This is a pre-change count, not measured post-change removal.

C Three geometries: old RTL and new RTL checked against an independent state model

Checked steps Accepted-write count Rejected-write count



3,023 steps | 2,411,234 word visits, each checked against both implementations | completion marker confirmed

Actual FF/LUT/timing savings remain unmeasured in this archive and require synthesis of the changed source snapshot.

Evidence levels are kept separate: mathematical invariant, baseline primitive instances, and finite functional regression. No cycle waveform is inferred.